

Project Design Phase-II

Technology Stack (Architecture & Stack)

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Team ID: VIPC2-MERN-CHETHULA HARISH

Project Name: Book A Doctor – Online Doctor Appointment Booking System

Technical Architecture

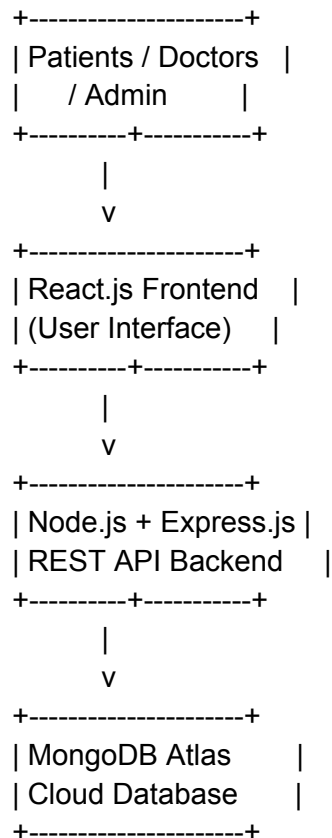


Table-1 : Components & Technologies

S.N o	Component	Description	Technology
1	User Interface	Web application interface for patients, doctors, and admins	React.js, HTML5, CSS3, JavaScript
2	Application Logic-1	User authentication and authorization	Node.js, Express.js
3	Application Logic-2	Appointment booking and management	Node.js, Express.js
4	Application Logic-3	Notification and doctor approval workflow	Node.js, Express.js
5	Database	Stores users, doctors, appointments, and notifications	MongoDB
6	Cloud Database	Cloud-hosted database service	MongoDB Atlas
7	File Storage	Stores application assets and documents	Local Storage / Cloud Storage
8	External API-1	REST API communication between frontend and backend	Axios API Calls
9	External API-2	Authentication token verification	JWT
10	Machine Learning Model	Not Applicable in current version	N/A
11	Infrastructure (Cloud)	Application deployment and hosting	Vercel (Frontend), Render (Backend)

Table-2 : Application Characteristics

S.N o	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for application development	React.js, Node.js, Express.js

2	Security Implementations	Password encryption, authentication, protected routes	JWT, bcryptjs
3	Scalable Architecture	Modular MERN architecture supports future expansion	MERN Stack
4	Availability	Cloud deployment ensures accessibility from anywhere	Vercel, Render, MongoDB Atlas
5	Performance	Fast API communication and optimized frontend rendering	React.js, Express.js, Axios

Architecture Benefits

- Secure user authentication and authorization.
 - Cloud-based database access.
 - Fast and responsive user experience.
 - Easy maintenance and scalability.
 - Supports future enhancements such as telemedicine and online payments.
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Conclusion

The Book A Doctor platform is built using a modern MERN Stack architecture that provides scalability, security, reliability, and high performance. The selected technology stack ensures efficient appointment management and supports future growth of the application.